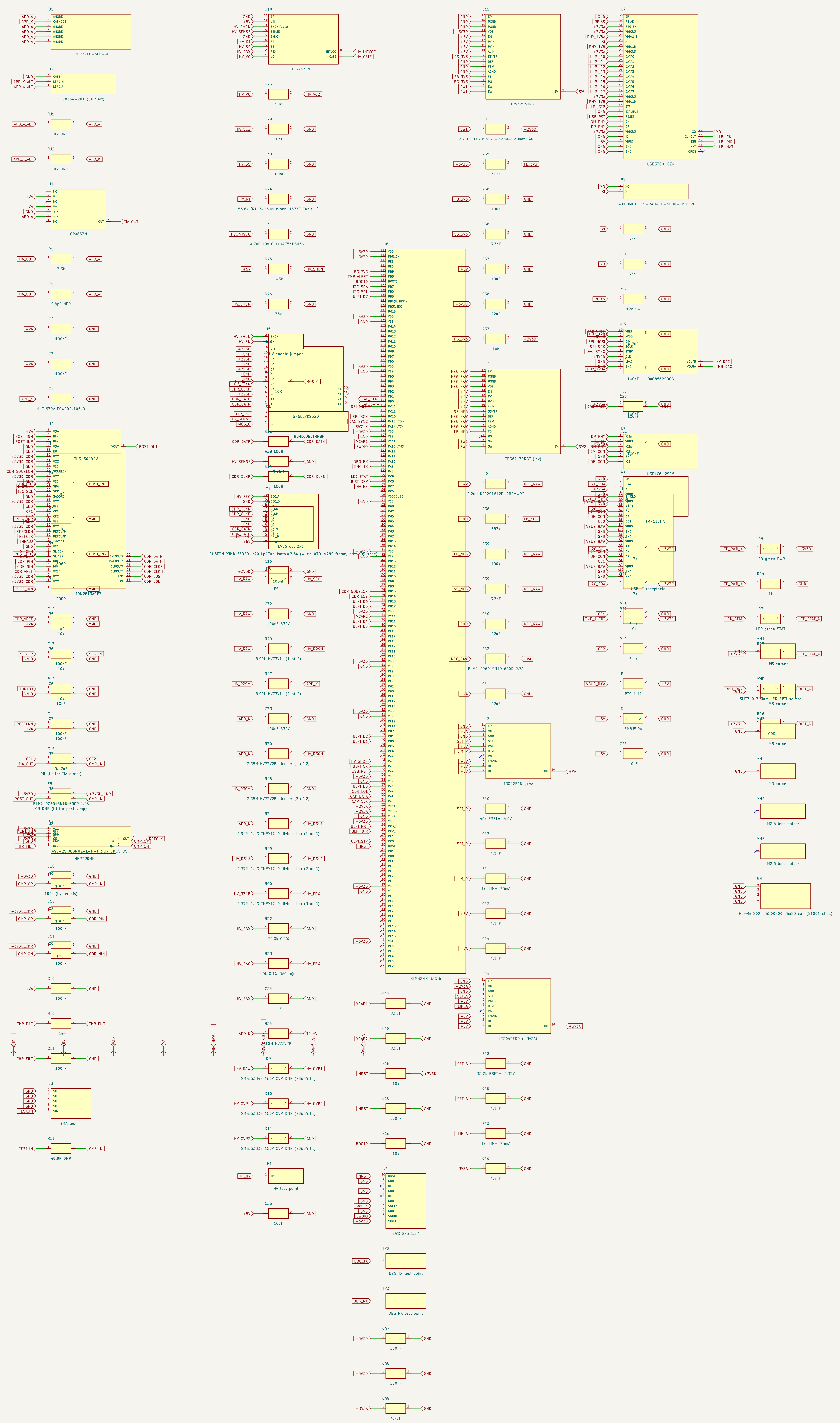
D1
C30737LH-500-90
D2
S8664-20K (DNP alt)
RJ1
0R DNP
RJ2
0R DNP
U1
OPA657N
R1
3.3k
C1
0.4pF NP0
C2
100nF
C3
100nF
C4
1uF 630V ECWFD2J105JB
U2
THS4304DBV
U4
ADN2813ACPZ
R2
1k
R3
806R
R4
200R
C12
R5
1uF
10k
C13
R6
100nF
10k
R12
C6
10k
10uF
C14
C7
100nF
100nF
C15
R7
0.47uF
0R (fit for TIA direct)
FB1
R8
BLM21PG601SN1D 600R 1.4A
0R DNP (fit for post-amp)
X2
U3
ASE-25.000MHZ-L-R-T 3.3V CMOS OSC
LMH7220MK
C28
R9
100nF
100k (hysteresis)
C50
C8
100nF
100nF
C51
C9
10uF
100nF
C10
100nF
R10
1k
C11
100nF
J3
SMA test in
R11
49.9R DNP
U10
LT3757EMSE
R23
10k
C29
10nF
C30
100nF
R24
53.6k (RT, f=250kHz per LT3757 Table 1)
C31
4.7uF 10V CL10J475KP8N3NC
R25
143k
R26
33k
J5
HV enable jumper
U5
SN65LVDS32D
R27
10R
Q1
IRLML0060TRPBF
R13
R28 100R
R14
0.05R
100R
T1
J2
LVDS out 2x3
CUSTOM WIND EFD20 1:20 Lp47uH Isat>=2.6A (Wurth 070-4290 frame, docs/09 spec)
C16
D5
100nF
ES1J
C32
100nF 630V
R29
5.00k HV73V1J (1 of 2)
R47
5.00k HV73V1J (2 of 2)
C33
100nF 630V
R30
2.35M HV73V2B bleeder (1 of 2)
R48
2.35M HV73V2B bleeder (2 of 2)
R31
2.94M 0.1% TNPV1210 divider top (1 of 3)
R49
2.37M 0.1% TNPV1210 divider top (2 of 3)
R50
2.37M 0.1% TNPV1210 divider top (3 of 3)
R32
75.0k 0.1%
R33
140k 0.1% DAC inject
C34
1nF
R34
10M HV73V2B
D9
SMBJ5384B 160V OVP DNP (S8664 fit)
D10
SMBJ5383B 150V OVP DNP (S8664 fit)
D11
SMBJ5383B 150V OVP DNP (S8664 fit)
TP1
HV test point
C35
10uF
U6
STM32H723ZGT6
C17
2.2uF
C18
2.2uF
R15
10k
C19
100nF
R16
10k
J4
SWD 2x5 1.27
TP2
DBG TX test point
TP3
DBG RX test point
C47
100nF
C48
100nF
C49
4.7uF
U11
TPS62130RGT
L1
2.2uH DFE201612E-2R2M=P2 Isat2.4A
R35
312k
R36
100k
C36
3.3nF
C37
10uF
C38
22uF
R37
10k
U12
TPS62130RGT (inv)
L2
2.2uH DFE201612E-2R2M=P2
R38
587k
R39
100k
C39
3.3nF
C40
22uF
FB2
BLM21SP601SN1D 600R 2.3A
C41
22uF
U13
LT3042EDD (+VA)
R40
46k RSET=+4.6V
C42
4.7uF
R41
1k ILIM=125mA
C43
4.7uF
C44
4.7uF
U14
LT3042EDD (+3V3A)
R42
33.2k RSET=+3.32V
C45
4.7uF
R43
1k ILIM=125mA
C46
4.7uF
U7
USB3300-EZK
X1
24.000MHz ECS-240-20-5PDN-TR CL20
C20
33pF
C21
33pF
R17
12k 1%
U8
C22
4.7uF
DAC8562SDGS
C23
100nF
C24
C26
100nF
100nF
D3
C27
100nF
USBLC6-2SC6
U9
J1
TMP117NAI
USB-C receptacle
R20
4.7k
R21
4.7k
R18
R22
5.1k
10k
R19
5.1k
F1
PTC 1.1A
D4
SMBJ5.0A
C25
10uF
D6
LED green PWR
R44
1k
D7
LED green STAT
MH1
R45
M3 corner
D8
MH2
SMT740 740nm LED BIST source
M3 corner
R46
MH3
100R
M3 corner
MH4
M3 corner
MH5
M2.5 lens holder
MH6
M2.5 lens holder
SH1
Harwin S02-25200300 25x20 can (S1001 clips)

Sheet: /
File: pdlvds_rx.kicad_sch

Title: PDLVDS-RX rev A — photodiode LVDS receiver

Size: A0
Date:
Rev: A
KiCad E.D.A. 8.0.9
Id: 1/1